JC2 H2 Prelim Paper 2 2024
Suggested Answers

Section A Cluster 4: Fieldwork

1 A group of 20 students wanted to investigate if the community is actively responding to climate change. They visited Tampines Town which was slated to be one of the eco-towns in Singapore, and did investigations on people's awareness and response to climate change actions in Tampines Hub.

They developed the following hypothesis: **The majority of the population in Tampines are aware of and are actively carrying out actions to mitigate climate change.**

The students went to Tampines Hub on a Saturday morning from 9am to 11 am, where the Hub is expected to be more crowded to get a bigger sample size. They went in groups of 5 all around Tampines Hub and approached any person who was willing to do their questionnaire.

They managed to survey 100 people in total, with 10 being in the age group of 51 and above and 2 of the respondents being Primary 5 students (See Resource 1) who were there on a visit to learn about the waste digester. They also took photographs of shoppers using the reverse vending machine in Tampines Hub and other eco-friendly facilities like the reusable bag station that were present in Tampines Hub.

Resource 1 shows pictures the students took of people engaged in green actions and facilities in Tampines Hub. Resource 2 shows the questionnaire survey that the students conducted. Resource 3A, 3B and 3C shows the results of the questionnaire that the student did to show their questionnaire findings.

(a) Explain one strength and one weakness of the student's hypothesis. [4]

- One **strength** is that the hypothesis is **measurable** as it seeks to investigate how aware and actively the population in Tampines is in carrying out actions to mitigate climate change [1]. Awareness can be determined through answer from the questions asked on climate change and actions can be observed in investigative fieldwork like what they see the population in Tampines doing. [1 additional mark]

- One **weakness** is that the hypothesis is **too generic** [1] and does not specify **what climate change actions are**, and if those actions carried out by the population actually mitigate climate change. [1 additional mark] ***OR*** One **weakness** is that the hypothesis is **double barrel**, and treats awareness and actions as one entity [1]. In reality, the population can be aware of climate change initiatives but not actively taking action to mitigate climate change. [1 additional mark]

- **Note: This is NOT weakness of hypothesis but weakness of the <u>investigation</u> per se**: The hypothesis states the majority of the population in Tampines are aware of and actively carry out climate change actions but the investigation is only for Tampines Hub.[X]

(b) Describe what you think the students were trying to show in each of the photographs they took. [5]

- According to Resource 1, the students were trying to show if the population in Tampines Hub were being made aware of and actively carrying out actions to mitigate climate change [1]
- In the 1st photo (top left), shoppers are seen taking a reusable bag that is being recycled again so that there is no need to use disposable plastic bags [1]
- In the 2nd photo (top right), the 2 students are seen on a visit to learn about the waste digester, so that they can be more aware of climate change strategies that are available [1]
- In the 3rd photo (bottom left), workers in the food court are seen handling waste, probably separating food waste from other waste for the waste digester to turn to fertilizers [1].
- In the 4th photo (bottom right), residents are seen using the reverse vending machine to recycle their empty containers [1].

(c) With reference to Resource 1, describe two possible issues related to research ethics that can arise from this investigation and how the students can overcome them. [4]

- One possible issue related to research ethics is that students **should not take photographs of persons without their permission** [1]. This can be resolved by asking for permission before taking photographs of persons in Tampines Hub. [1 additional mark]

- Another possible issue related to research ethics is that they **should not be doing surveys in the mall without permission from the mall management**. This can be an issue when shop owners complain that the students are interfering into their business by approaching shoppers [1]. This can be resolved by asking for permission to do surveys from the mall management before they embark on their investigation and maybe have a tag that says they have been approved to do surveys. [1 additional mark]

- Another possible issue related to research ethics is that they **should not enter work areas** like in photograph 3 (bottom left) where they might **disturb work processes or even compromise safety** as they are not authorised to enter these places [1]. This can be resolved by not entering these areas and instead request for photographs of climate change actions within the development from the management of the mall or the owner of the food court instead. [1 additional mark]

(d) Suggest how the students could minimise potential risks while carrying out their questionnaire survey (Resource 2) at Tampines Hub. [6]

- Risk of **angry shoppers**– as some shoppers who are unsure of the legitimacy of the students doing a survey or might not want to be interrupted while shopping, the students run the risk of an angry shopper shouting at them or being unreasonable [1]. Thus students should approach the management at Tampines Hub first to get

permission to do the survey in the mall – preferably with some identification tag to proof their legitimacy. [1 additional mark]

- Students also face the risk of **physical danger** when they enter places of production that they are not supposed to, like into the area in photograph 3 (bottom left). These places might have machinery that can be dangerous when in operation [1]. Students should just keep away from these areas that they are unauthorised to enter. [1 additional mark]

- Another risk the students might face is **theft** since they are doing fieldwork in a busy shopping area, especially when they have their valuables with them [1]. The students should keep all their valuables (like handphones, wallets) out of sight in their bags or pockets so that they will not be victims of theft while they are focused on doing their fieldwork. [1 additional mark]

- [also can, but not the best] Students also face the risk of **getting lost** in Tampines Hub as it is a big area while they are doing their survey and walking around in groups of 5 [1]. They should either go around in smaller groups (of 2) or ensure that they have each other's contact and not get lost. [1 additional mark]

(e) With reference to Resources 1, 3A and 3B, explain the importance of community participation in climate change initiatives. [6]

- Community participation is important because collaborative partnerships allow each stakeholder to **bring** their unique expertise, resources, and capabilities **to the table** [1]

- In R1, it is seen that **raising awareness** by partners like businesses is important as it inspires climate action (eg. recycling and waste reduction) [1]. Businesses also provides **platforms** and **funding** for start-ups and initiatives like the use of food waste digesters (R1) that children can learn about. [1 additional mark]

- In R3A, it is seen that **participation by residents and home owners** are important also as their **actions and habits** can contribute to positive climate change actions [1]. It is seen that most of those surveyed use the blue bin for recycling at least 3-4 times a week which **collectively can reduce waste** in the long term. [1 additional mark] ***OR*** R3A shows that more can use less air-conditioning because more than 80 of them use air-conditioning 3-4 times a week, which can add to more heat being generated back to the environment and **more CO2 emission**. [1 additional mark]

- In R3B, it is seen that **awareness of** and **community participation and support** of climate change initiatives are important for positive climate change action as only then can the change **begin** [1]. R3B shows that more persons are aware of the recycling facilities in Tampines, and they also support the use of reusable bags when they go shopping, which can lead to a **reduction of waste**, especially one time use plastics. [1 additional mark] ***OR*** R3B shows that not many are aware of where community gardens are in Tampines, which can **reduce the effectiveness** of community participation in greening efforts. [1 additional mark]

(f) With reference to Resources 3A, 3B and 3C, explain the strengths and limitations of the data presentation methods the students have used. [6]

- R3A **gives a clear indication** of the climate change actions/ inactions that are being practiced (eg. the use of the blue recycling bins and the frequency of use of air-conditioning) [1]. The use of bar graphs allows the presentation to **make comparison** between the two question, Q1 and Q3. [1 additional mark]

- However, Q1 and Q3 are **opposing questions** where a higher frequency in the use of air-conditioning is not good while a higher frequency in use of blue bin is good [1]. This can be confusing when one is trying to see the trend in climate change actions. [1 additional mark]

- R3B also shows a similar use of bar graphs for YES/ NO answers on awareness of climate change facilities and habits that **clearly shows** the response of shoppers [1]. Placing both answers (YES/ NO) side by side to each question allows one to **compare** which answer is higher and also which initiative is more popular amongst the population. [1 additional mark]

- However, there are **no numbers** that indicate how many shoppers said YES and how many said NO. [1] Therefore, the reader has to **second guess or estimate** the number based on the lines on the graph, making it cumbersome and error prone. [1 additional mark]

- R3C gives a clear indication of the modes of transport of all the respondents and it also clearly show **how many (person/ percentage count) per each slice** of the pie chart [1]. This is especially useful when the total number of respondents is also 100, but even if it is not, the pie chart is **useful** for representation of the percentages of **big numbers** of more than 100. [1 additional mark]

(g) Based on observations at Tampines Hub (Resource 1) and results from the questionnaire survey, the students concluded that their hypothesis is correct. Evaluate the validity of their conclusion. [10]

- Hypothesis: ***The majority of the population in Tampines are aware of and are actively carrying out actions to mitigate climate change.***

The students' hypothesis is ***somewhat VALID***:

- From the **observations** the students made at Tampines Hub (R1), it can be seen that there are initiatives of climate change actions, and the population seems to be **aware and participate** in it. From young students (photograph 2, top right) to adults (photograph 1 and 4, top left and bottom right) to workers (photograph 3, bottom left), there seems to be enough awareness of climate change actions in Tampines, making the observations seem **accurate** in validating the students' hypothesis.

- The questions asked in the **questionnaire** (R2) are also relatively **encompassing**. The students do consider other climate change mitigation actions like use of air-conditioning and infrastructure like community gardens besides just recycling.

- The respondents in Tampines Hub also **seem to be aware** of what climate change actions are and do participate in it (R3A and 3B), like the bringing of reusable bags and the awareness of recycling facilities within the Hub. The respondents in Tampines Hub also **seem to be actively carrying our actions** to mitigate climate change by taking public transport (R3C). However, the use of public transport might not be a deliberate move towards climate change mitigation, but could simply be because they do not own a car due to the high prices of cars in Singapore or the convenience of Tampines Hub by public transport. Thus, this question is **not indicative** of positive climate change action.

However the conclusion is also ***NOT fully VALID***:

- Hypothesis is on the majority of the population in Tampines, but the survey is **only done in Tampines Hub**, for only 2 hours on a Saturday morning. Although they had 100 respondents, their responses might **not be representative** of the majority of the population in Tampines as the **sample is limited** to those who go shopping on a Monday morning. Those who work on Saturday would probably not have made this group, and their response would be important to the data set as well. Students **should** instead go on 2 separate days and may once in the morning and once in the afternoon to ensure that they get a greater variety of samples and a greater sample size. They **should** also do their questionnaire in other places in Tampines to ensure a greater coverage of the population.

- Their questionnaire with **only 6 questions** (R2) is **not reliable** in measuring climate change actions as it only covers specific actions which might not lead to mitigation of climate change, as mitigation of climate change do encompass other areas – eg. food waste, type of foods consumed, etc. that is not reflected in the short questionnaire.

- The **actions** of the population seems sporadic and is **not reliable** to indicate that they are actively carrying out climate change actions to mitigate climate change. Eg. Majority of the population say they do use a reusable bag for shopping and are aware of the recycling facilities in Tampines Hub (R3B), but are not aware of community gardens in Tampines (R3B).

Section B Cluster 3: Sustainable Future and Climate Change

2. Resource 4 shows the distribution of urban liveability of Chinese cities. Resource 5 shows the frequency of hot days in Chinese cities. Resource 6 is a photograph of a farmer using a hose to water plants near a dying chilli pepper plant at his farm in Longquan village in south-western China's Chongqing municipality on Aug 20, 2022.

(a) Describe the distribution of urban liveability of Chinese cities as shown in Resource 4. [4]

Award 1 mark for each description of the urban liveability of Chinese cities, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

- The **eastern coastal regions** of China, particularly around major cities like Beijing, Shanghai, and Guangzhou, are marked with high to very high liveability. [1]
- **Beijing**, the capital of China also has very high urban liveability. [1]
- Surrounding the areas of high liveability, there are regions marked as having medium liveability.
- The **central and western regions of China**, including the Northwestern and Southwestern regions, are predominantly marked with low to very low liveability. [1]

(b) Suggest reasons why urban liveability is very high in the selected cities shown in Resource 4. [6]

Award 1 mark for each suggested reason why urban liveability is very high in the selected cities, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each suggestion, where applicable.

Possible responses include:

Urban liveability in cities like Beijing, Shanghai, Chongqing, and Shenzhen is very high due to several key factors:

- These cities are **economic powerhouses**, with high GDP per capita and abundant employment opportunities, particularly in finance, technology, manufacturing, and service industries. [1] This economic strength translates into better living standards, high quality public services, and well-maintained infrastructure. [1 additional mark]

- Beijing and Shanghai, in particular, are **hubs for international business**, attracting multinational corporations and offering a globalized environment that enhances economic opportunities and quality of life. [1]

- These cities boast **well-developed transportation systems**, including extensive metro networks, high-speed rail links, and efficient airports. [1] This connectivity enhances mobility, reduces commute times, and contributes to overall urban liveability. [1 additional mark]

- Ongoing **urban development projects**, such as **smart city initiatives** in Shenzhen, have significantly improved infrastructure, integrating technology to enhance urban management and the quality of services provided to residents. [1]

- Beijing and Shanghai are home to some of **China's top universities and research institutions**, offering high-quality education and attracting students and professionals from across the country and internationally. [1]

- These cities provide access to **top-tier medical facilities and healthcare services**, which contribute to higher life expectancy and overall well-being. [1]

- The presence of **large expatriate communities** in these cities, particularly in Shanghai and Beijing, creates a multicultural environment that enhances social vibrancy and global connectivity. [1]

- The central and local governments have implemented **comprehensive urban planning and governance strategies** that prioritize liveability. [1] For example, in Shenzhen, the government's focus on innovation and technology has spurred the city's rapid development into a modern, livable metropolis. [1 additional mark]

- Shenzhen and Beijing are at the forefront of **integrating technology into urban management through smart city initiatives**, which improve everything from traffic management to energy efficiency, making the cities more liveable. [1]

(c) Describe the distribution of hot days in Chinese cities as shown in Resource 5. [4]

Award 1 mark for each description of the distribution of hot days in Chinese cities, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

- The image provided shows a color-coded map of China, indicating the distribution of hot days across different regions.
- The highest concentration of hot days is predominantly in the **southeastern part of China**. [1]
- Surrounding the areas with the highest concentration, there is a significant region marked in red. This area appears to cover a **large part of eastern China**. [1]
- Medium frequency as shown in orange are spread across **central**, **and some parts of eastern China**. [1] **Chongqing** also experiences medium frequency of hot days. [1 additional mark]
- The areas with low frequency of hot days (light orange) are **scattered across all the regions,** but with large areas in the north and northwestern part of China. [1]
- The areas with very low frequency of hot days (yellow) are mainly found in the **northeast, north and southwestern parts of China**. [1]

(d) Explain how the urban liveability of Central and East China may be affected by the number of hot days as shown in Resource 5. [6]

Award 1 mark for an explanation of how urban liveability of Central and East China may be affected by the number of hot says, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:

- As temperatures rise, **heat-related illnesses** such as heatstroke, dehydration, and respiratory issues are likely to increase. [1] **Vulnerable populations, including the elderly and children, are particularly at risk**. [1 additional mark] Higher temperatures can also **exacerbate chronic conditions** like cardiovascular and respiratory diseases, reducing overall public health and increasing healthcare costs. [1 additional mark]

- High temperatures can **decrease worker productivity**, particularly in sectors requiring outdoor labour, such as construction and agriculture. [1] Prolonged exposure to heat can lead to **fatigue and heat exhaustion**, resulting in lower efficiency and increased absenteeism. This can nega**tively affect the economic output** of cities in these regions. [1 additional mark]

- With more hot days, there will be a **greater demand for air conditioning** and cooling systems, leading to **increased energy consumption**. [1] This can **strain the electrical grid**, potentially causing power outages. [1 additional mark] Additionally, the increased energy demand can contribute to higher greenhouse gas emissions if the energy is sourced from fossil fuels, **worsening urban air quality and climate change impacts**. [1 additional mark]

- Hotter temperatures can lead to increased evaporation rates, **reducing the availability of surface water and groundwater resources**. [1] This can **exacerbate water scarcity issues**, especially in urban areas where demand is high. [1 additional mark]

- Increased temperatures can lead to the **deterioration of green spaces and agricultural land**, affecting the local environment and biodiversity. [1] The loss of vegetation can further contribute to the **urban heat island effect**, creating a feedback loop that worsens living conditions in these regions. [1 additional mark]

(e) With reference to Resource 6, describe the impact of climate change on agricultural ecosystems in China. [4]

Award 1 mark for each description of the impact of climate change on agricultural ecosystems in China, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

- The dry, withered appearance of the plants suggests they are **suffering from heat stress**, which is a common consequence of rising temperatures due to climate change. [1] High temperatures can lead to **reduced crop yields and even crop failure**. [1 additional mark]

- The farmer is manually watering the plants, indicating that **natural water sources may be insufficient**. [1] Drought conditions, exacerbated by climate change, can reduce water availability for irrigation, leading to **water scarcity in agricultural areas**. [1 additional mark]

- The dying chili pepper plant indicates that **crop productivity is being compromised**. [1] Extreme weather events such as prolonged heatwaves and droughts can lead to a decline in agricultural output, **affecting food security**. [1 additional mark]

- The need for the farmer to use a hose to water the plants suggests an **increased demand for irrigation to counteract the effects of drought**. [1] This can put additional **pressure on water resources**, particularly in regions already facing water scarcity. [1 additional mark]

- Climate change can **alter the timing and length of growing seasons**, making traditional farming practices less effective. [1] This may require farmers to **adjust planting schedules or switch to more resilient crop varieties**. [1 additional mark]

(f) Suggest adaptation strategies that China can implement to cope with the impact of climate change on agricultural ecosystems. [6]

Award 1 mark for each suggested adaptation strategy that China can implement to cope with the impact of climate change on agricultural ecosystems, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each suggestion, where applicable.

Possible responses include:

To cope with the impact of climate change on agricultural ecosystems, China can implement several adaptation strategies that address both short-term vulnerabilities and long-term sustainability:

- China can promote the **development and adoption of drought-resistant, heat-tolerant, and saline-resistant crop varieties**. [1] These crops can withstand extreme weather conditions, ensuring food security even during adverse climate events. [1 additional mark]

- Encouraging farmers to **grow a variety of crops** can reduce the risk of total crop failure due to climate change. [1] Crop rotation and intercropping can also improve soil health and reduce pest outbreaks. [1 additional mark]

- **Investing in water-saving technologies** such as drip irrigation, sprinklers, and rainwater harvesting can optimise water use in agriculture, particularly in regions facing water scarcity due to climate change. [1]

- **Implementing wastewater recycling for agricultural use** can provide an additional water source while reducing the environmental impact of water use in agriculture. [1]

- **Building infrastructure that can withstand extreme weather events**, such as climate-resilient roads, storage facilities, and greenhouses, can help protect agricultural outputs and ensure continuous production. [1]

- **Expanding access to renewable energy in rural areas** can support the adoption of modern farming techniques and reduce the carbon footprint of agriculture. [1]

- **Investing in research on climate change impacts and adaptation strategies** specific to China's diverse agro-ecological zones can provide farmers with the knowledge and tools needed to adapt. [1]

- **Engaging local communities** in the planning and implementation of adaptation strategies ensures that the solutions are culturally appropriate and address the specific needs of the population. [1]

Section C

3. 'Strategies to meet the needs of the elderly living in cities are often sufficient in addressing the economic aspects only.' Evaluate this statement. [20]

Possible Approaches:

Candidates could approach the question by making a judgement on whether strategies to meet the needs of the elderly living in cities are only sufficient in addressing the **economic** aspects, while **other aspects such as social well-being and psychological well-being** are not sufficiently addressed. The strategies could be evaluated in relation to criteria such as the **strengths and limitations**, **scale of impact** of the strategy and whether it has addressed the **root causes** of the problem faced by the elderly. Candidates could also analyse the **unintended impacts** of the strategies.

Candidates could also approach the question by making a judgment on whether strategies to meet the needs of the elderly living in cities are only sufficient in addressing the economic aspects in some contexts than others through a consideration of **two or more case studies.** Candidates could analyse the **contextual factors such as the involvement of different stakeholders, political will, economic viability, resources available**, which may have contributed to the differences in the sufficiency of strategies to meet the varied needs of the elderly. Candidates could also analyse the **challenges**, which are common across the case studies.

Levels marked using Generic Level Descriptors for H2 essays

4. Evaluate the roles of key actors in climate change response. [20]

Possible Approaches:

Candidates could approach the question by making a judgement on the relative significance of the roles of key actors in climate change response. Candidates could consider the roles of key actors such as the state, NGO and firms in climate change response. Candidates can analyse the **necessity and importance of their roles** by considering the **degree/ extent to which they can exert an influence** on climate change actions, and **how they can influence each other** in decision-making.

Candidates could also approach the question by making a judgement on the **different roles played by key actors in climate change response in different contexts**. This can be done through a consideration of **two or more case studies** where the roles were different due to local variations in relation to economic, social, physical and political background. Candidates could analyse the **contextual factors such as level of economic development, effective institutions and governance and physical environment**, which may have contributed to the differences in the relative significance of the roles.

Levels marked using Generic Level Descriptors for H2 essays